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Instruction Manual for DMFC Utility Tool

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Hitachi Metals Ltd.

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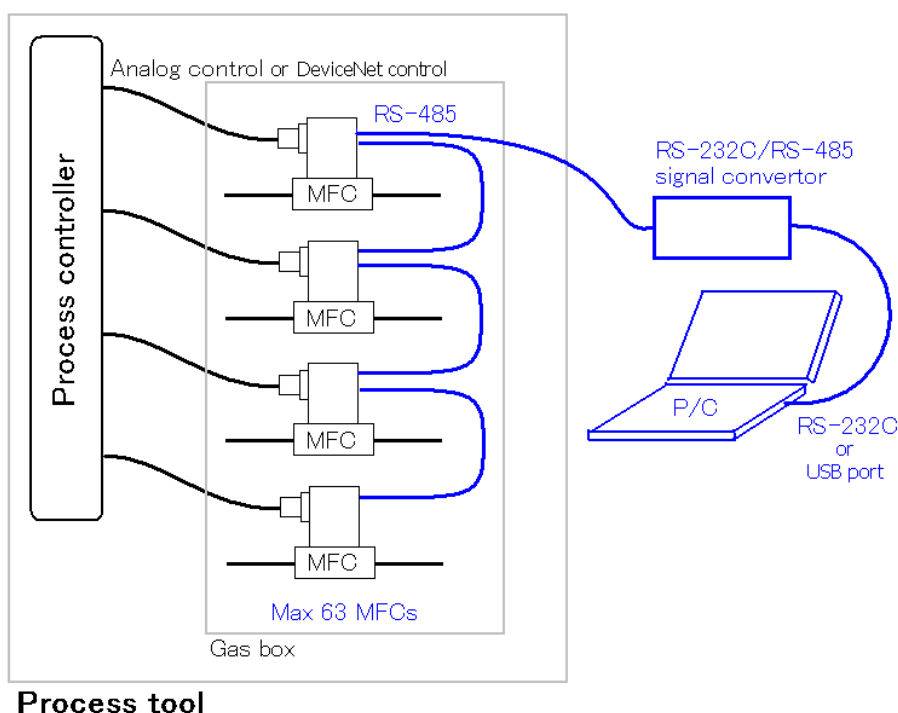
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1 Overview

This application, which is designed for Hitachi Metals digital mass flow controllers (hereafter called DMFC), enables the following functions via RS-485 serial communication manipulated from computer: Changing of setpoint (at digital control only), response waveform monitoring, valve control (opening/closing), comment entry on individual DMFC, changing of MFC ID and baud rate, zero adjustment, and alarm setting. Some DMFCs are capable of checking delivery inspection data in DMFC and acquiring information through the zero-point record control function.

If a DMFC-laden process controller exercises analog or DeviceNet control of MFC, this application allows saving of acquired data in files as well as simultaneous monitoring of DMFC control information, including setpoints, outputs and valve voltages, of multiple DMFCs, assuring no effect on process. With the system in digital control mode, this application enables the control of DMFC flow rate and the acquisition of corresponding response waveform data, which ensures easy DMFC operation check. The reading and control of zero-point record data stored in DMFC are available if DMFC is endowed with the zero-point record control function, which is of use for MFC preventive maintenance.



Process tool

2 Features

Key features of this application include:

- Monitoring of the items as follows as to all the connected DMFCs: DMFC ID number, serial number, process gas name, full-scale flow rate, conversion factor, control mode (Digital or Analog control), current setpoint, current output, current valve voltage, and status (presence of errors)
- Setting of DMFC ID number
- Setting of DMFC serial communication speed (baud rate)
- DMFC control mode selection (Digital or Analog control)
- Error/alarm recovery in DMFC
- Comment entry on individual DMFC (up to 40 characters), comment access
- Batch auto zero adjustment of individual DMFC or all the connected DMFCs
- Entry of flow rate setting, monitoring of DMFC output and valve voltage, and data sending to MS Excel™ by this application at Digital control
- For DMFC outfitted with the calibration data save function, invocation and sending (to MS Excel™) of calibration data containing flow setting data, N2 measured data, response time, factory-shipped zero-point, and pre-shipment inspection data including valve leakage
- For DMFC outfitted with the zero-point record control function, reading of zero-point data [%] and cumulative zero-point shift amount [%] relative to the default, and reset of record data

3 PC System Requirements

This section describes the PC system requirements to use this application.

3.1 Computer

- Model: PC/AT-compatible computer (RS232C interface-laden computer. A computer with no RS232C port may become available under the condition that a USB serial conversion adapter is used. COM number should be max. COM 10 regardless of computer type.)
- Display: 848 x 593 pixel resolution
1024 x 768 pixel resolution with 256 colors or more is recommended.
- Operating systems: Windows 98, Windows 2000, Windows XP, Windows Vista, Windows 7
- Memory: This application requires 256MB of RAM or better.
- Drives: A hard drive with at least 10MB of free space, and besides another free disk space for data saving.

3.2 RS-232C/RS-485 Communication Converter

RS-232C serial communication is capitalized on in this application to establish communication with DMFC. Conversion from RS-232C to RS-485 requires the use of the RS-232C/RS-485 communication converter because of RS-485 communications protocol on DMFC. Our converter CS-485 (COM port connection) or USB485-AE1 (USB port high-speed connection) is available. USB485-AE1 comes with a USB cable (1.5m) and modular cable (1m).



Caution

CS-485 converter supports communication speed, up to 38.4kbps.
For communication at higher baud rate, use the signal converters listed below.

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B&B electronics
Bits

BSU485-AE1 (USB port connection type)
485BAT3 (COM port connection type)
BRS-485/Z1 (COM port connection type)

3.3 RS232C Communication Cable

RS-232C straight type cable is required for connection between the computer and RS232C/RS485 signal converter. The cable is to be connected with all the pins.

3.4 Modular Cable

A modular cable is used for connection between RS232C/RS485 signal converter and DMFC.

RJ-11 6-pole straight type cable is required, and can be used if connected with Pins 3 and 4.

Our cables listed below are available.

Model	Cable length
LC-04D	1 m
LC-08D	3 m
LC-05D	5 m
LC-06D	10 m

4 Preparation

4.1 Installation

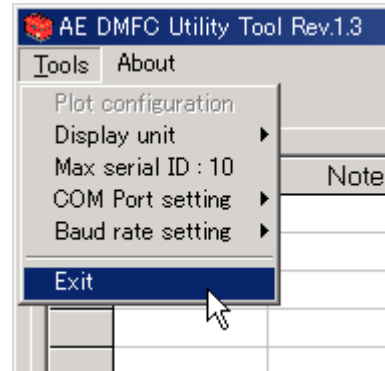
Place this CD-ROM into your CD-ROM drive. Double-click on the setup.exe, and follow the on-screen instructions to complete the installation.

Note: Be sure to log in with Administrator authority if your system is Windows 2000 or Windows XP Professional.

4.2 Program Start and Exit

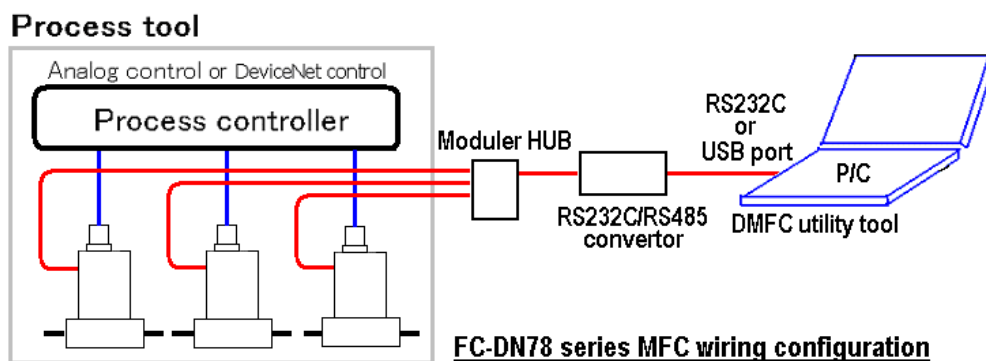
To start the program: Click on the DMFC utility tool icon from Start menu.

To exit the program: Click on Exit from Tools menu on the main window. (See right)

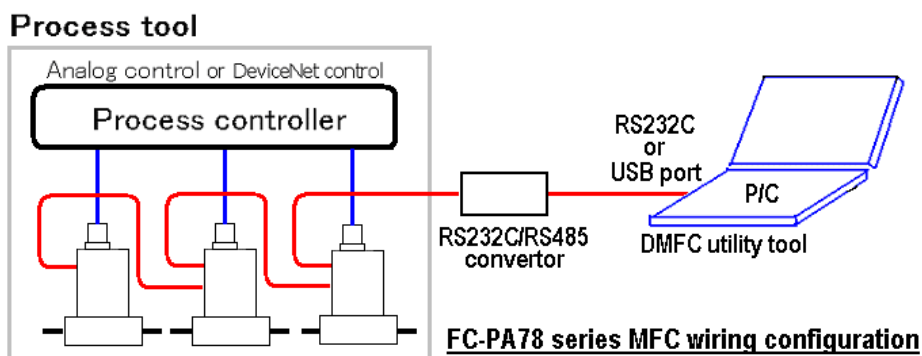


4.3 Connection Diagram (Examples)

4.3.1 When the modular connector has a single port in DMFC (FC-DN Series MFC)



4.3.2 When the modular connector has 2 ports in DMFC (FC-PA Series MFC)



The following connection cables should conform to the required length: RS-232C cable (or USB cable) between the computer and RS-232C/RS-485 signal converter; Max. 5m, and RS-485 modular cable between RS-232C/RS-485 signal converter and DMFC; Max. 50m.

4.4 Other Notes

1. Acceptable Use Policy for Excel

This application permits acquired data to be transferred to Microsoft Excel™ via DDE communication. Basic precautions should always be followed for safe use, including the following:

- Do not open more than one Excel file. Otherwise, this application fails to recognize a desired file.
- A worksheet is to be named “Sheet*” (* is assigned with a digit such as 1, 2, 3).

2. Flow control Mode

Non-DeviceNet DMFC

Our DMFC allows you to choose the type of flow control, analog control or digital control. Analog control is a default upon power supply to DMFC.

[Analog control mode]

Analog control is to control a flow rate through the adoption of 0 to 5VDC analog voltage from the external device. MFC setpoints issued with digital command is overridden if the system is in analog mode. As with digital control mode, this mode is capable of digital communication except for flow control.

[Digital control mode]

Digital control executes digital command via serial communication for flow control, which causes analog flow setting to be overridden.



Warning

Digital control takes effect only when DMFCs, except some models, receive the digital control switch command through digital communication, which causes the flow setting from the analog connector to be overridden and allows flow control with the command through digital communication. **Note that DMFC operation from the system becomes disabled if switching of flow control from Analog to Digital by this application is attempted while DMFC is in process through analog voltage.** If temporary switching of flow control to Digital is performed by this application to acquire data on response waveforms, be sure to place DMFC in a proper control mode such as switching to Analog control.

DeviceNet DMFC

No analog control mode is available in this type of DMFC. DeviceNet DMFC allows flow control in response to DeviceNet communication command or RS485 communication command. Only setting command from DeviceNet is effective for flow control upon power supply to DMFC, which causes serial communication command to be overridden. Once DeviceNet DMFC is connected with this application, flow control through DeviceNet setting command and serial communication command become enabled.

3. MFC ID Overlap

This application enables the simultaneous connection of DMFCs, up to 63 if an individual ID number is assigned to each DMFC. ID numbers for all DMFCs are factory-configured to “7”. Initial use of this application requires single connection to DMFC. Then, network connection is to be established according to section **4.3 Connection** with unique MFC ID numbers assigned.



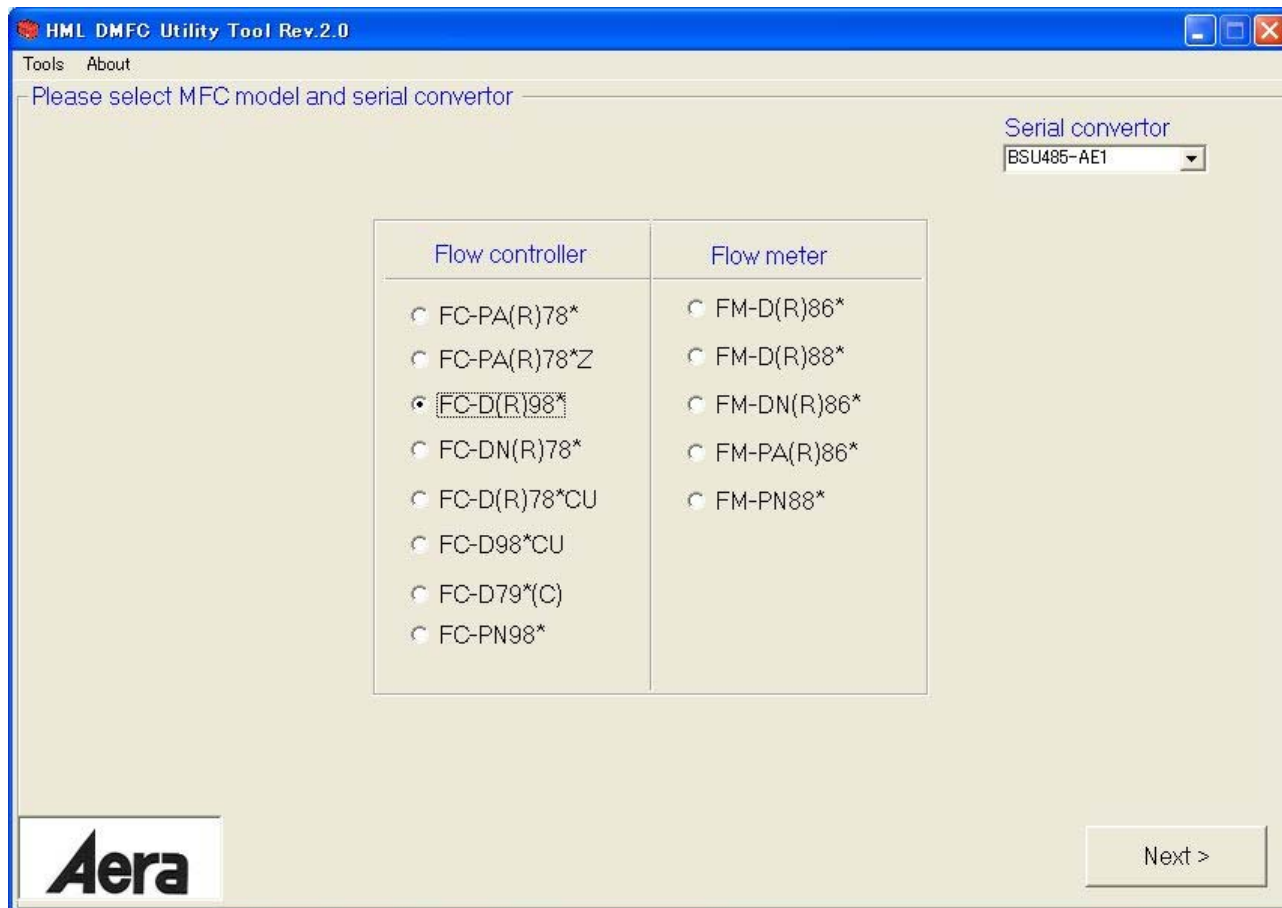
Warning

Abnormal operation is observed if this application detects an overlap between ID numbers.

5 Operation Screens

5.1 Screen Immediately after Program Start

The screen displayed immediately after program start allows you to select not only DMFC model for communication but RS-232C/RS485 signal converter type. The functions in this application and baud rate become selectable if any DMFC model is selected. Once RS-232C/RS485 signal converter type is specified, data is stored in this application that requires no further type selection.



Next > button at the lower right of the screen becomes available if DMFC model and RS-232C/RS485 signal converter type are specified. This button is used to proceed to the following screen.



Caution

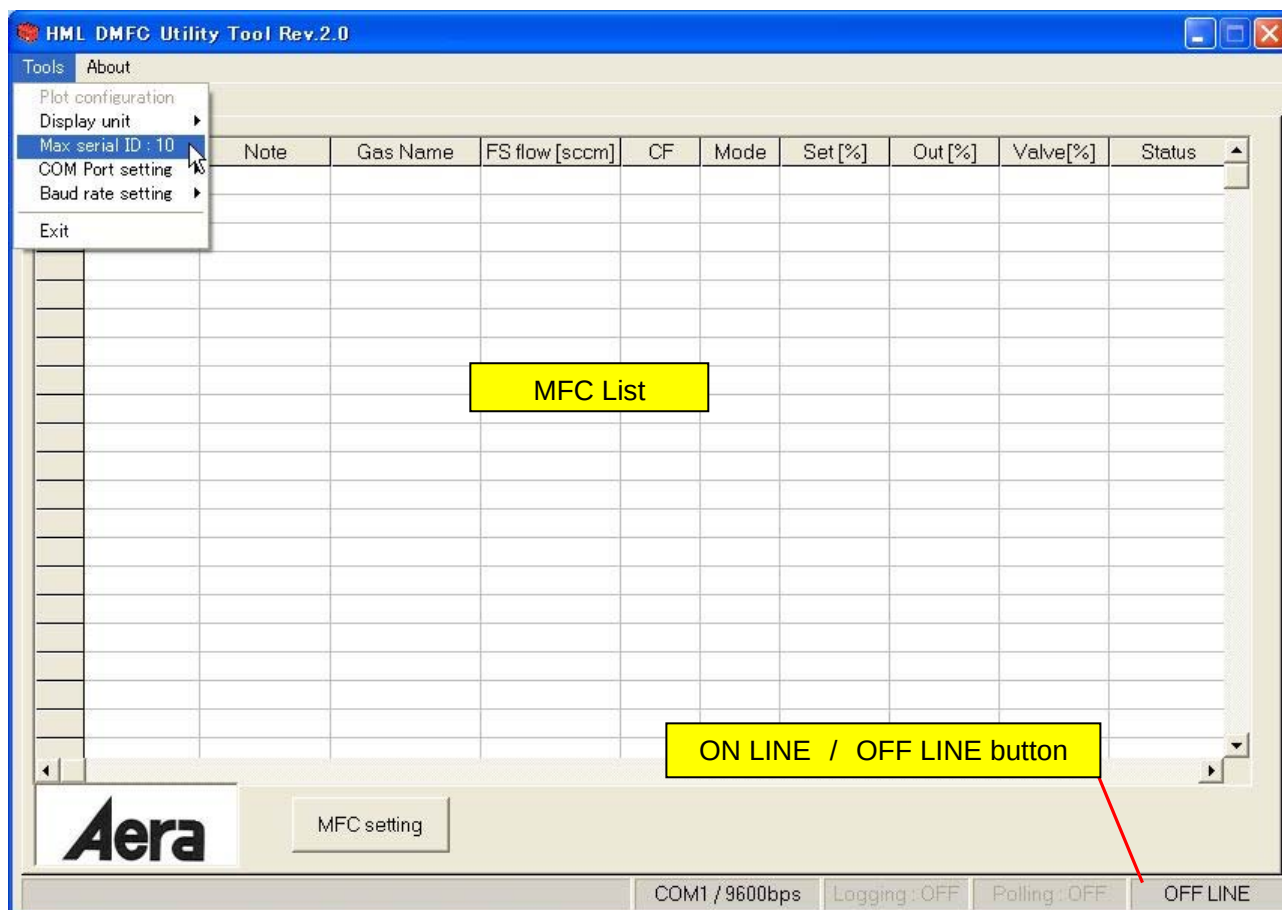
Contact Hitachi Metals Ltd. if DMFC to connect this application to carries multiple models.

5.2 Select MFC Screen

5.2.1 Communication start/stop

From Tools menu, check the following three items prior to commencing communication with DMFC.

- ① Is proper Max serial ID assigned? The default is 10 that permits DMFCs assigned with ID numbers 1 to 10 to be connected with. Connection to DMFCs with ID numbers 11 and higher is disregarded. For communication to DMFCs with ID numbers 11 and up, click on Max serial ID menu and set an ID number larger than the default.
- ② Is proper COM port number assigned? → See **6.3 COM Port Setting**.
- ③ Is proper baud rate set? → See **6.4 Baud Rate Setting**.

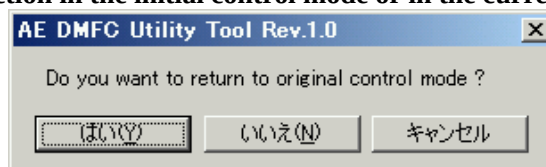


OFF LINE button, located at the lower right of the screen, is used to change the serial communication condition between connected and disconnected. To commence communication with DMFC, click on **OFF LINE** button. **OFF LINE** denotes that communication is disconnected (button shown in gray), and **ON LINE** denotes that it is connected (button shown in green). Communication disconnection is an example given on the above screen.



Caution

This application is capable of storing the control modes of all the DMFCs upon connection. If disconnection is attempted in a switched control mode while this application is in run, the following screen is displayed to pose disconnection choice, either disconnection in the initial control mode or in the current mode.



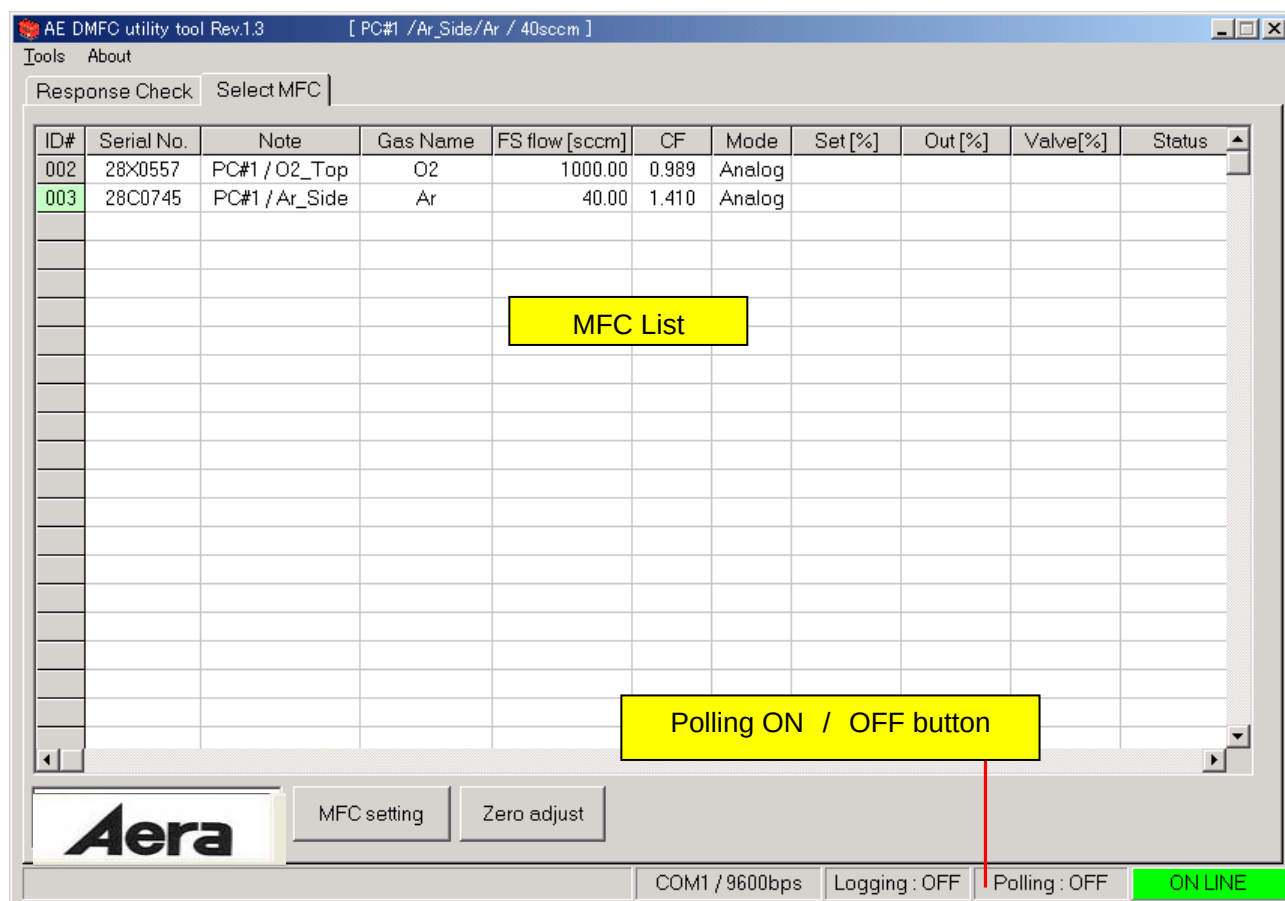
To return to the initial control mode, click on the “Yes”. To proceed with disconnection in the current mode, click on the “No”.

Click on **ON LINE** button to terminate the connection, which switches the screen to one displayed immediately after program start.

Information on the relevant DMFC is listed (MFC List) if DMFC connection completion is assured.

ID#	ID number (changeable)
Serial No.	MFC serial number (non-changeable)
Note	Notes recorded in MFC (Default: No record, changeable)
Gas Name	Process gas name (non-changeable)
FS flow	Full-scale flow rate (non-changeable)
CF	Process gas conversion factor (non-changeable)
Mode	Control mode (changeable between analog and digital)
Set[%]	Flow setpoint
Out[%]	Flow output
Valve[%]	Valve voltage
Status	DMFC status including errors and alarms

The following screen is displayed if DMFCs with ID numbers 2 to 3 are connected. In this example, data on DMFC with ID number 003 is highlighted in light green. DMFC highlighted is regarded as a primary, active MFC. Data on the current active MFC is also provided on the title bar on this screen. A multitude of functions, such as acquiring response waveform data and zero-point adjustment and control, are available in the active MFC. DMFC with the largest ID number is considered an active MFC if multiple DMFCs are recognized at the time of the initial connection. With the click of the name of any desired DMFC in MFC List, the intended DMFC is assigned for active MFC.

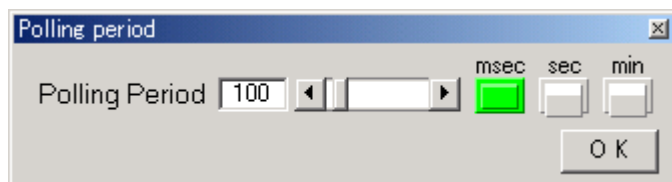


See chapter 7 “Troubleshooting” if the desired DMFC fails to be recognized.

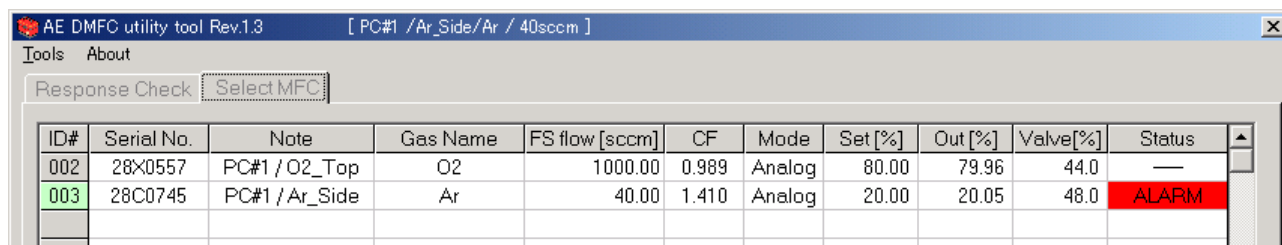
5.2.2 Polling Function

This function is to ensure continuous monitoring of the following data at established intervals as to all the connected DMFCs: flow setpoint, flow output, valve voltage, and DMFC status.

Polling time interval is to be set before performing polling. With the right click of **Polling: OFF** button located at the lower right of the screen, the following dialog box is displayed. Set arbitrary time intervals with the use of the slider and time unit buttons. Polling time interval setting takes effect with the click of **OK** button.



The button is displayed in green when **Polling: OFF** button at the lower right of the screen is clicked, which prompts continuous monitoring of the current flow setpoint, flow output, valve voltage, and status of the intended. An actual polling time interval is shown on **Polling** button.

A screenshot of the "AE DMFC utility tool Rev.1.3" window. The title bar shows "[PC#1 / Ar_Side/Ar / 40sccm]". Below the title bar are "Tools" and "About" menus. A "Response Check" button is visible, followed by a "Select MFC" button. Below these is a table with 11 columns: ID#, Serial No., Note, Gas Name, FS flow [sccm], CF, Mode, Set [%], Out [%], Valve [%], and Status. The table has two rows of data. The second row (ID# 003) is highlighted in red, and its "Status" cell contains the word "ALARM" in red text.

ID#	Serial No.	Note	Gas Name	FS flow [sccm]	CF	Mode	Set [%]	Out [%]	Valve [%]	Status
002	28X0557	PC#1 / O2_Top	O2	1000.00	0.989	Analog	80.00	79.96	44.0	—
003	28C0745	PC#1 / Ar_Side	Ar	40.00	1.410	Analog	20.00	20.05	48.0	ALARM

In the example given above, the two connected DMFCs control the flow rate through the adoption of analog voltage (set at 80% in O2 MFC and 20% in Ar MFC). The above screen indicates that an error has been raised in Ar MFC. Access to details on errors and alarms is available on the MFC setting window for checking.



Caution

If DMFC 7 or earlier version is polled in analog control mode, the setpoint and valve voltage are expressed only in the unit of 1% (MFC control assured with 0.006-%FS resolution (14 bits)). The valve voltage is to be updated once every second. DMFC version is presented in "Version" on the MFC setting window.

5.2.3 Zero-point adjustment during polling

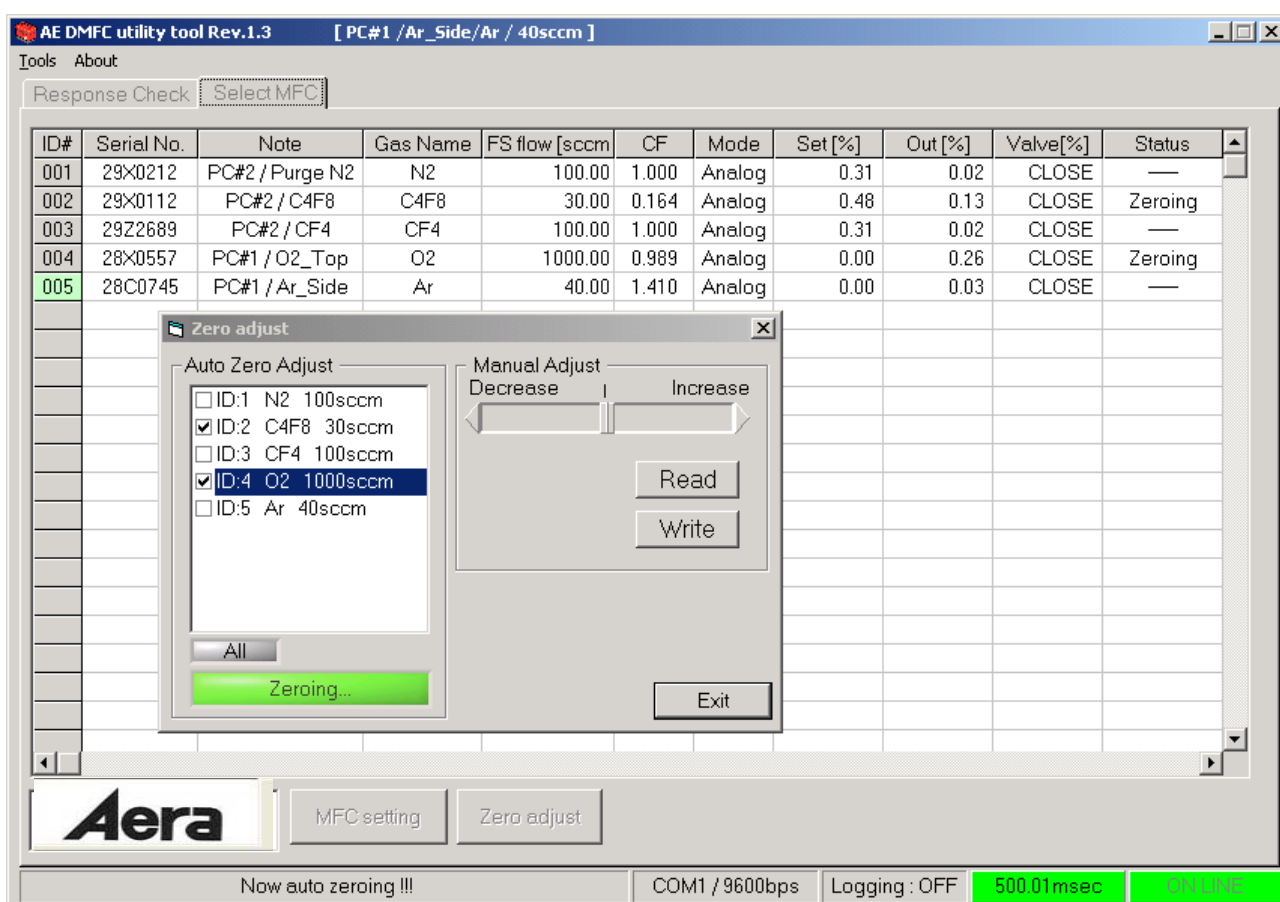


Caution

The following requirements should always be satisfied before performing zero-point adjustment.

1. DMFC flow setpoint is set at 0%.
2. Complete shutoff of the gas is assured.
The valve provided in DMFC is not a full shutoff valve. Ensure that the shutoff valve located at the DMFC upstream or downstream shuts off gas flow completely.
3. Adequate warming up of DMFC is provided.
Refer to the relevant DMFC specification for warming-up time.

With the click of **Auto zero** button while DMFC is being polled, the Zero Adjust window shown below is displayed. The window enables you to make zero adjustment of the active MFC.



The method of zero adjustment falls into the following two types:

1. Manual zero adjustment

The Manual Adjust function enables manual adjustment of zero point of the active MFC to any desired value. This function is of use in zero-point offset adjustment if the process controller fails to read a negative voltage.

- **Slider** : This bar is used to change a zero adjustment value.
- **Read** button : This button is used to read the current zero adjustment value of the active DMFC. As DMFC Ver8 or earlier, the slider is reset to the center with the click of **Read** button in Fine setting.
- **Write** button : This button is used to send a zero adjustment value changed with the slider to DMFC.

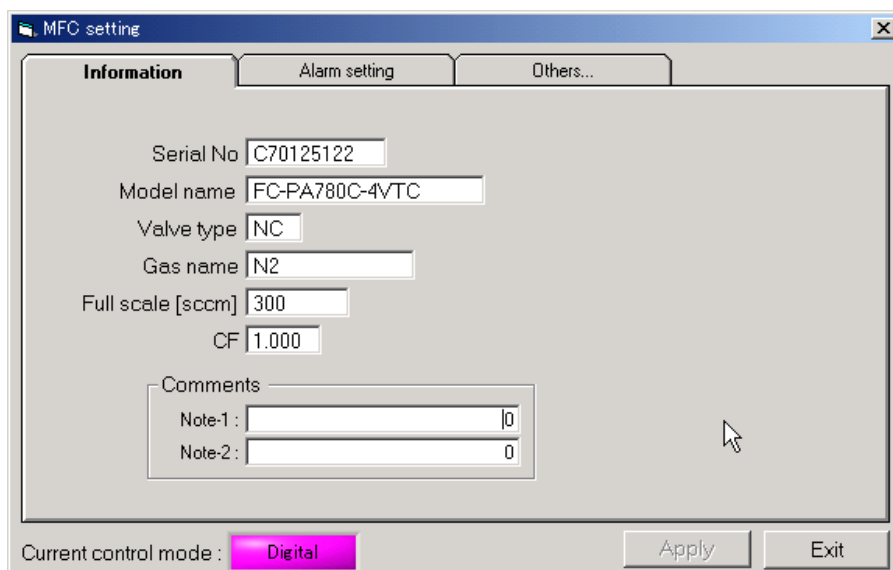
2. Automatic zero adjustment

This button is used to conduct auto zero-point adjustment of DMFC. With the checkbox of an intended DMFC marked in the list, “Zeroing” is displayed in Status with the click of Auto zero button. Auto adjustment requires approx. 10 to 30 seconds, and “Complete” is displayed upon completion of adjustment. “Error” denotes adjustment error due to some reason. Auto zero-point adjustment is performed to DMFCs with ID numbers 2 and 4 in the example provided in the previous page. If DMFC is endowed with the zero-point record control function, DMFC energization accumulated time, pre-adjustment zero-point data and cumulative zero-point shift amount relative to the default are to be stored in DMFC.

5.2.4 MFC setting function

With the click of the MFC Setting button, the current values of the active MFC specification and alarm function are displayed.

Information tab



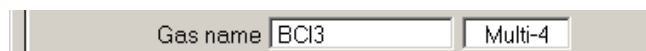
The screenshot shows the 'MFC setting' window with the 'Information' tab selected. The fields are as follows:

Field	Value
Serial No	C70125122
Model name	FC-PA780C-4VTC
Valve type	NC
Gas name	N2
Full scale [sccm]	300
CF	1.000
Comments	Note-1: , Note-2:

At the bottom, the 'Current control mode' is set to 'Digital' (highlighted in pink). There are 'Apply' and 'Exit' buttons.

- Serial No MFC serial number
- Model name Model name (DMFC 7 or later version)
- Valve Valve type, normally-closed (NC) or normally-open (NO)
- Gas name Process gas name
- Full scale Full-scale flow rate
- CF Process gas conversion factor
- Comment Entry of two types of 20-character comment (1-byte alphanumeric) can be stored in DMFC EEPROM.
- Current control mode This button is used to select the DMFC control mode between analog and digital. Available in non-DeviceNet DMFCs.

In case of MGMR(multi) MFC, flow range BIN size will be displayed at right side of Gas name

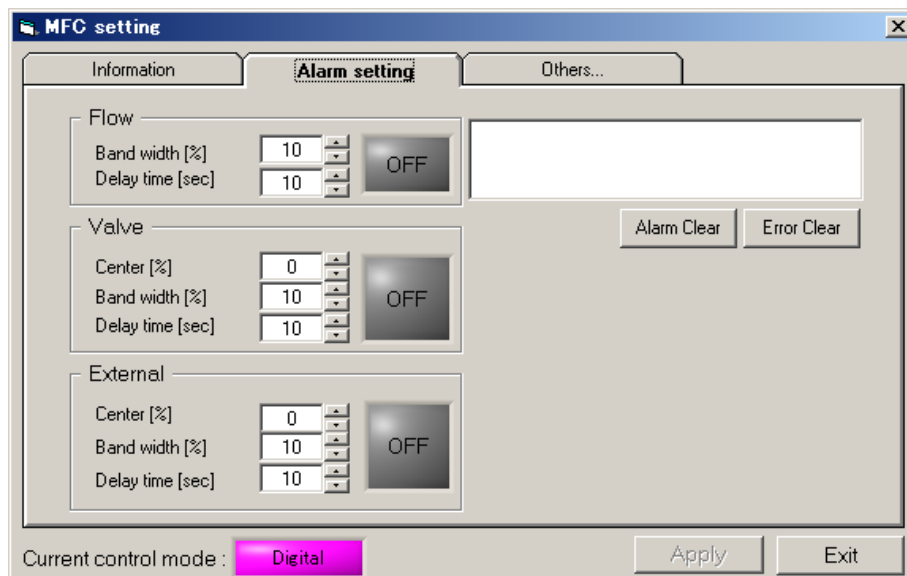


The screenshot shows the 'Gas name' field with the value 'BCI3' and a dropdown menu showing 'Multi-4'.

Alarm setting tab

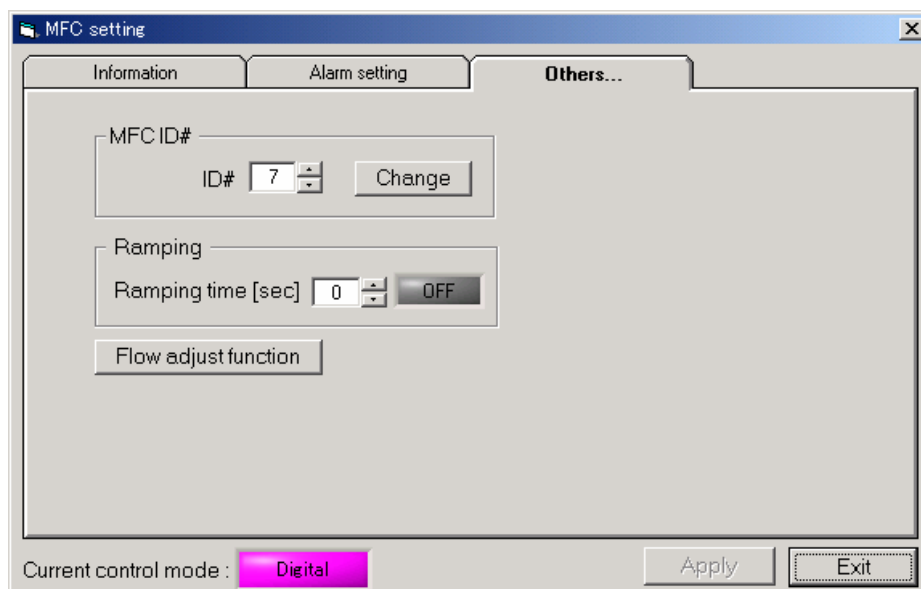
Non-DeviceNet DMFC permits programming of 3 types of alarms as follows: Flow alarm, Valve alarm, and External alarm.

Refer to the relevant DMFC Instruction Manual for details.



- **Alarm Clear** button This button is used to clear an alarm occurred.
- **Error Clear** button This button is used to clear an error occurred.

Others... tab



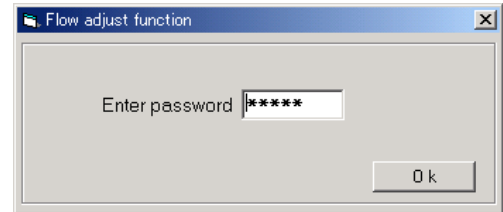
- **MFC ID#** This is the current MFC ID number. After setting a new MFC ID number, changes take effect with the click of the **Change** button.
- **Ramping** The default ramping time is 0. Changes take effect with the click of the **Apply** button if ramping time is altered. The Ramping function is disabled if set to "0". Ramping time is to be set in the 0- to 999-second range. If ramping time is set at 5sec, ramping control takes place allowing 5-second shift from the current setpoint to the next setpoint. This is effective to avoid abrupt change in gas flow.

- Flow adjust function

This function is used to adjust flow gain value. The flow rate can be increased / decreased with this function. Also, it is possible to return to original gain value (except PrimAera series, FC-D98*CU, FC-PN98*, FM-PN88*).

Normally, this function is not available and password-protected since this function effects flow rate directly. Please ask our sales section when you use this function.

With the click of **Flow adjust function** button, the following dialog box is displayed. Please ask our sales section and we'll show the password.



With correct password, the following dialog box is displayed. Flow rate adjustment can be done by inputting "Gain adjust" value then clicking the **Apply** button. The range of "Gain adjust" value that you can apply is 0.920 to 1.080 (=1.000+/-0.080).

For example, the initial value is "0.952" as following window.

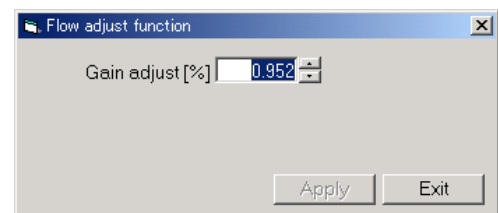
To obtain -1.5% lower flow, the new gain adjust value is calculated as follows.

$$0.952 - 1.5/100 = 0.952 - 0.015 = 0.937$$

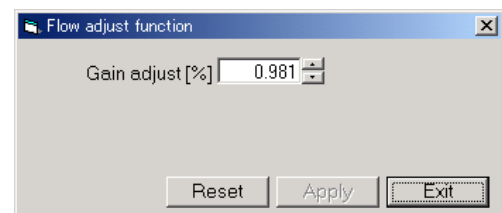
To obtain +2.9% higher flow,

$$0.952 + 2.9/100 = 0.952 + 0.029 = 0.981$$

The new value is available after clicking the **Apply** button.

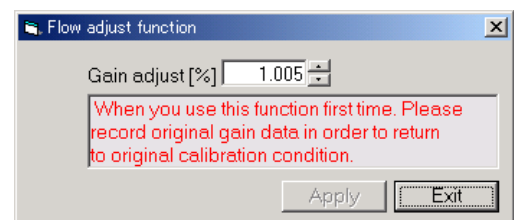


The **Reset** button is displayed as following figure when the initial value is changed. It is possible to return to the original value with the button clicking.



In case of PrimAera series(FC-D98*CU, FC-PN98*, FM-PN88*), the following message is displayed in the dialog box.

The function that can return to the original value is not available in this case. The original value will be overwrote by new value after clicking the **Apply** button. When you use this function first time, please record original value in order to return to original value.



Special setting tab

In case of connected MFC is FC-D780CU or FC-D780CUY, Special setting tab is appeared as below. MAC ID value and/or default control mode can be set. Click Apply button to send setting value to the MFC after changing.

The screenshot shows the 'MFC setting' dialog box with the 'Special setting' tab selected. The dialog has four tabs: 'Information', 'Alarm setting', 'Others...', and 'Special setting'. The 'Special setting' tab contains two dropdown menus: 'MAC ID set' with the value '0x21(33)' and 'Default Control Mode' with the value 'Digital'. At the bottom, there is a status bar showing 'Current control mode : Digital' and two buttons: 'Apply' and 'Exit'.

In case of connected MFC is DeviceNet MFC, Special setting tab is appeared as below. Input/Output instance value can be set. Click Apply button to send setting value to the MFC after changing.

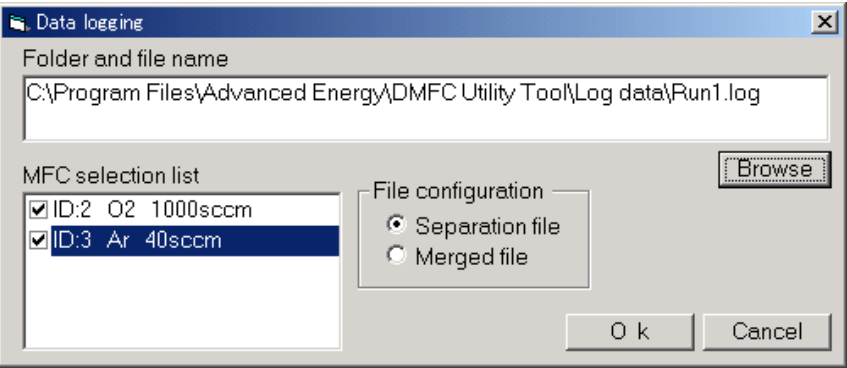
The screenshot shows the 'MFC setting' dialog box with the 'Special setting' tab selected. The dialog has four tabs: 'Information', 'Alarm setting', 'Others...', and 'Special setting'. The 'Special setting' tab contains four input fields: 'DeviceNet serial No.' with the value '00737149', 'Input instance' with the value '2', 'Output instance' with the value '7', and 'Full scale (hex)' with the value '6000'. At the bottom, there is a status bar showing 'Current control mode : Digital' and two buttons: 'Apply' and 'Exit'.

5.2.5 Data logging function

The DMFC flow setpoint, flow output, and valve voltage can be saved as a log file while DMFC is being polled. Log files are a text file with the setpoints, outputs and voltages separated by commas (“,”), which can be derived in Excel format.

Data logging

The following window is displayed with the click of **Logging: OFF** button.



Folder and name
MFC selection list
File configuration
 Separation file

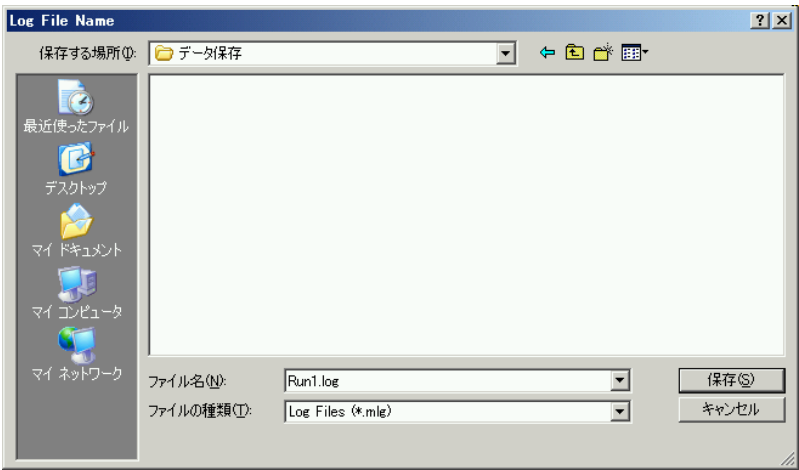
The names of the folder and file in which a log file is stored
DMFC whose checkbox is marked is to be logged.

 Merged file

A DMFC-to-DMFC log file is created if this checkbox is marked. The file is recorded under a filename that is designated in “Folder and name” with “_DMFC serial number” tacked to its end. As the designated filename is “Run1.log” in the above example, the file is recorded under the filename of “Run1_C050604653.log” if DMFC is assigned with serial number C050604653. Data on DMFC that is selected in MFC Selection List is saved in one log file if this checkbox is marked. A filename, which is designated in “Folder and name”, is to be allocated to the file.

Browse button

This button is used to display the dialog box shown below.
Enter a folder name to save a log file in and log file name.



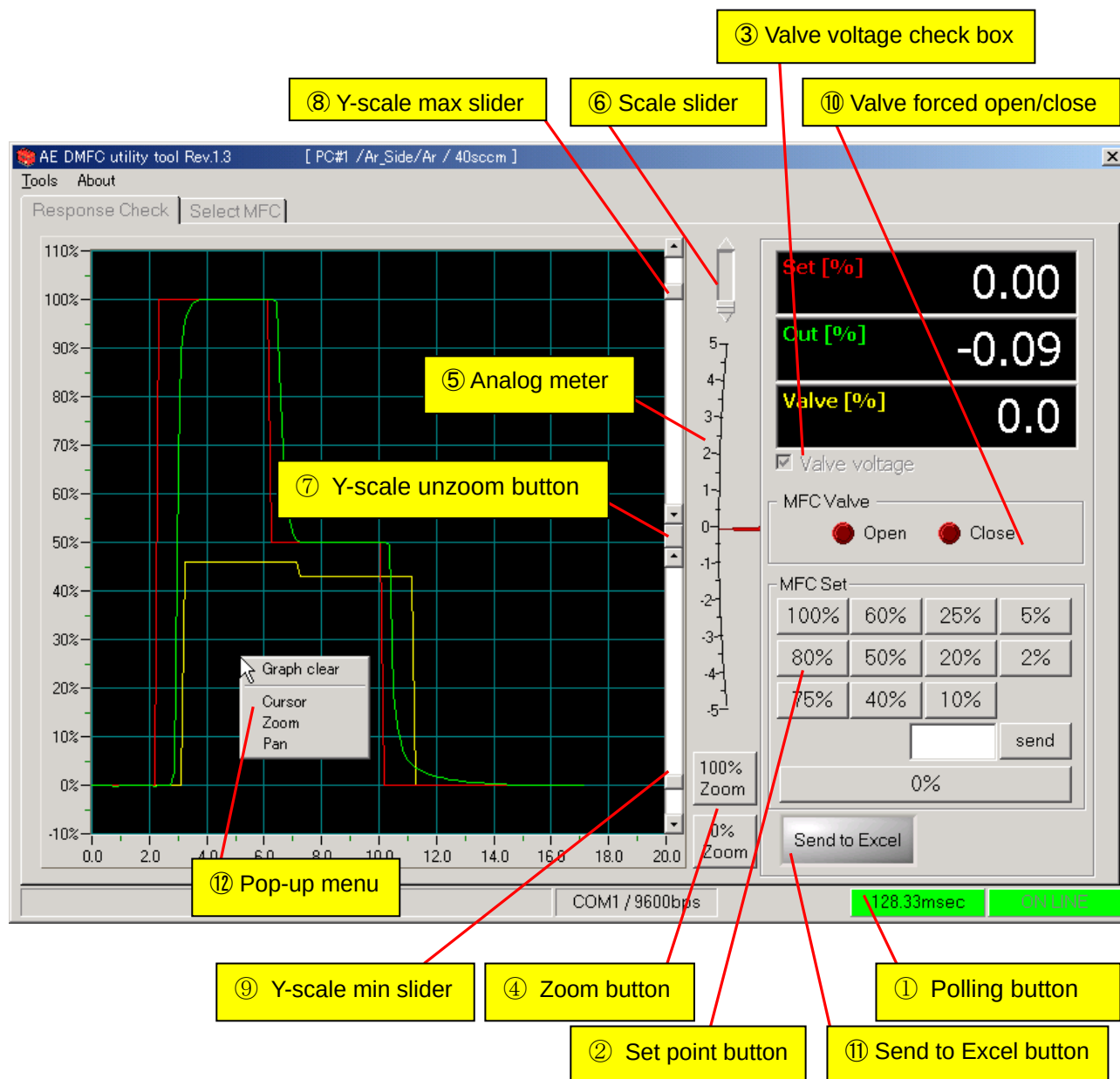
Cancel button
OK button

This button is used to cancel data logging.
This button is used to start data logging, and it is changed to **Logging: ON** button upon initiation of data logging. The instant when **OK** button is pressed is deemed as the starting point of data logging.

The polling function enables log data containing DMFC data to be saved. The log files, which are insusceptible to the cancellation of polling, remain open and stand ready to allow data acquisition. To end data logging, click on **Logging: ON** button and close the log file.

5.3 Response Check Screen

This screen enables checking of response waveforms of the active MFC. A desired DMFC needs to be designated as an active MFC on the Select MFC screen prior to calling up this screen.



① Polling button

This button is used to not only conduct continuous acquisition of active MFC data containing setpoint and outputs but also present it in graphical waveform. To cancel polling, re-click on this button. Actual polling intervals are presented on **Polling** button during waveform acquisition. With waveform acquisition in process, the right click of a mouse causes scrolling on the graph to pause. See “6.1 Setting of Sampling Time” for sampling time setting.

② Set point button

This button is available only when the active MFC is under Digital control. With click of any of the buttons, the relevant setpoint is sent to DMFC. To assign a desired setpoint, enter the intended setpoint in the textbox provided on the right of “0%” button and click on **Send** button. The entered setpoint is to be sent to DMFC upon click of **Send** button.

③ **Valve voltage check box**

The initial settings of graphical waveforms include a flow setpoint (green line) and flow output (red line) only. If this checkbox is marked, an obtained valve voltage is presented in yellow line on the screen.

④ **Zoom button**

The 100% Zoom button is used to establish a range of the graph screen specified with the analog meter (⑤) centering on a 100% basis. The 0% Zoom button is used to establish the range specified with the analog meter centering on a 0% basis.

⑤ **Analog meter**

This meter reads the outputs of the active MFC.

⑥ **Scale slider**

This bar is used to change the upper/lower limit of the analog meter (⑤) scale.

⑦ **Y-scale unzoom button**

This button is used to reset the vertical axis scale on the graph to the minimum value (-10%) and maximum value (+110%).

⑧ **Y-scale max slider**

This slider is used to change the maximum value of the vertical axis on the graph. The Y-axis maximum value increases with the upward shift of the slider, and it decreases with the downward shift.

⑨ **Y-scale min slider**

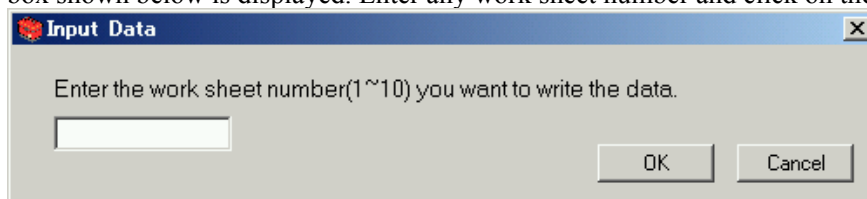
This slider is used to change the minimum value of the vertical axis on the graph. The Y-axis minimum value increases with the upward shift of the slider, and it decreases with the downward shift.

⑩ **Valve forced open/close**

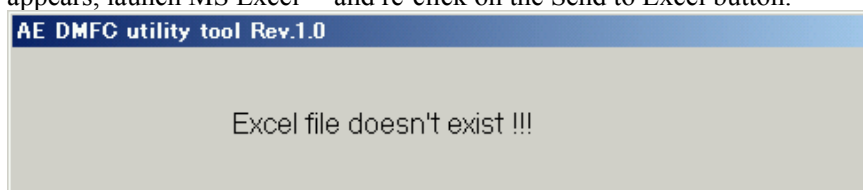
This button is used to permit the forced opening and closing of valves.

⑪ **Send to Excel button**

This button is used to send waveform data to MS Excel™. With the click of this button, the dialog box shown below is displayed. Enter any work sheet number and click on the OK button.



The following message is displayed if MS Excel™ is not activated. In the event that the message appears, launch MS Excel™ and re-click on the Send to Excel button.



Example: When waveform data is sent to MS Excel™

	A	B	C	D	E	F
1						
2	0.05	0	-0.04			
3	0.1	0	-0.04			
4	0.15	0	-0.04			
5	0.2	0	-0.04			
6	0.25	0	-0.04			
7	0.3	0	-0.04			
8	0.35	0	-0.04			
9	0.4	0	-0.04			
10	0.45	0	-0.04			
11	0.5	0	-0.04			
12	0.55	0	-0.04			
13	0.6	0	-0.04			

(From Column A) Time [sec], setpoint [%] and output [%]
Enter any item names in the first row that is blank.

⑭ Pop-up menu

The pop-up menu appears with the right-click of the graph, which offers you the option of the four functions listed below:

Graph clear: This is used to erase the graph.

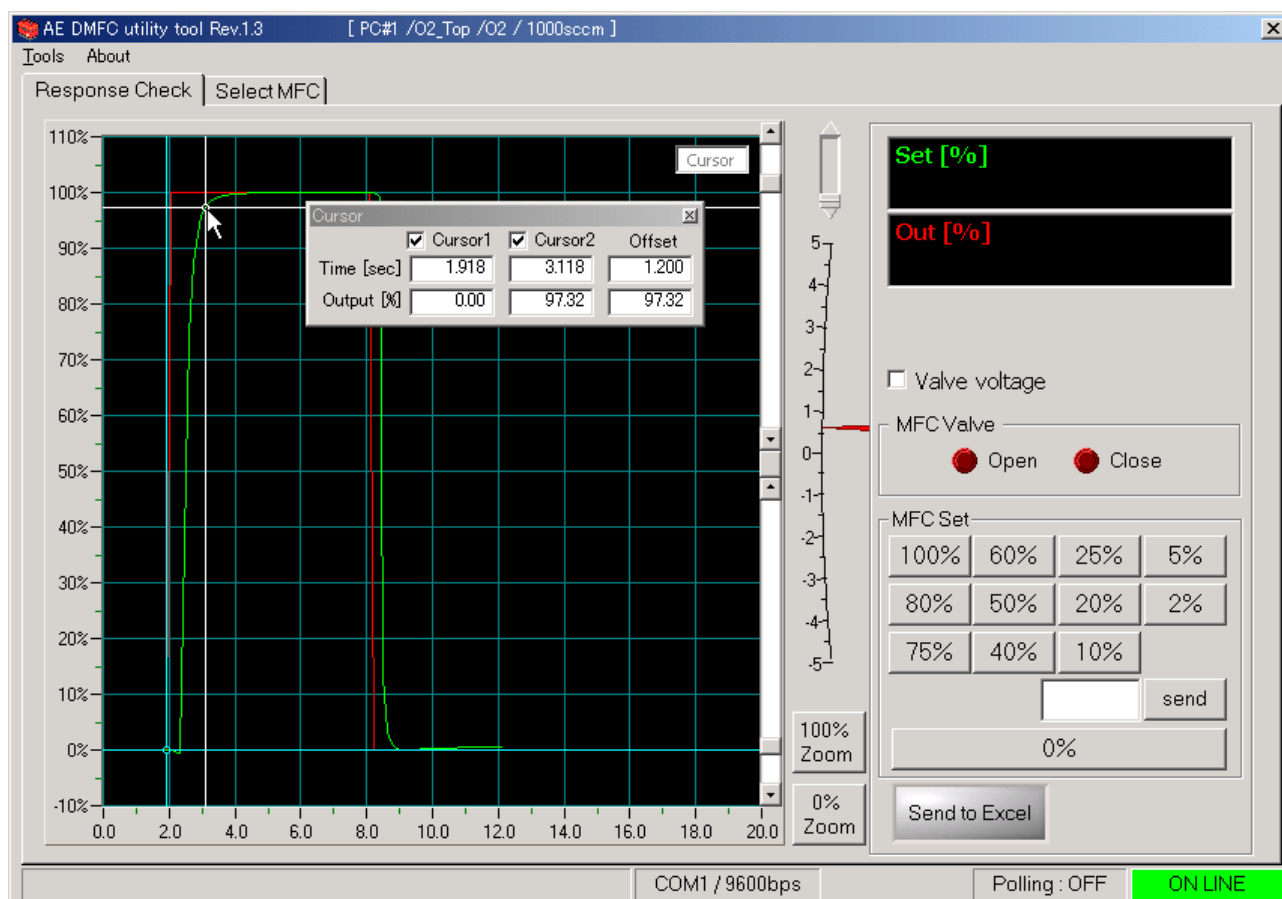
Cursor: Cursor

Zoom: Screen zooming

Pan: Screen panning

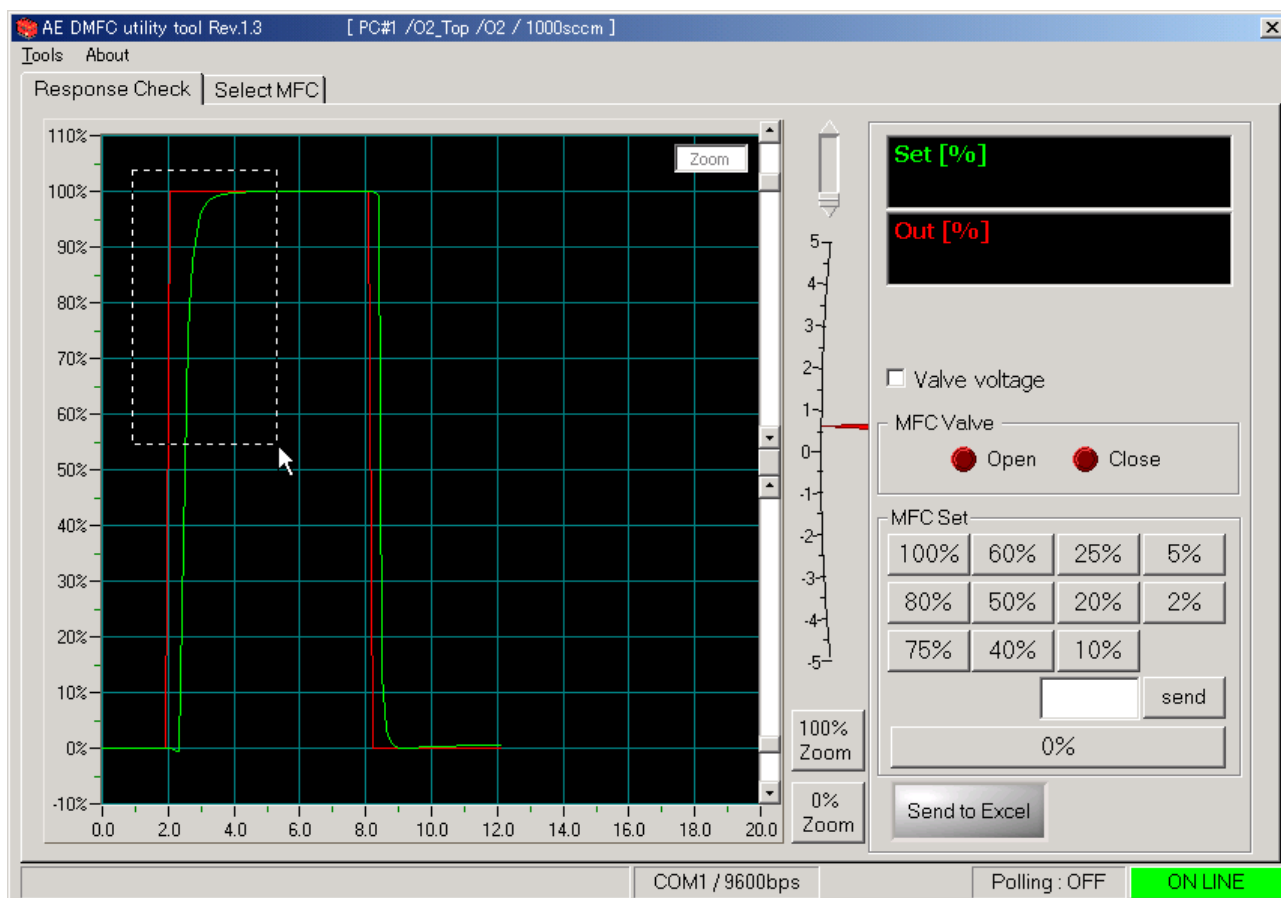
Cursor

The two cursors, in cyan (Cursor1) and in white (Cursor2), and window are displayed as shown below if “Cursor” is selected from the menu. Time and MFC output are displayed on the window as the two cursors dragged with a mouse-click, which provides the difference between two cursors.



Zoom

With “Zoom” selected from the menu, a waveform is scaled up through dragging and dropping on the graph.



Pan

With “Pan” selected from the menu, a waveform is panned from right to left or up and down, through dragging and dropping on the graph.



Caution

If DMFC 7 or earlier version is polled in analog control mode, the setpoint and valve voltage are expressed only in the unit of 1% (MFC control assured with 0.006-%FS resolution (14 bits)). The valve voltage is to be updated once every second. DMFC version is presented in “Version” on the MFC setting window.

5.4 Certification Data Screen

This screen is displayed only if FC-PA78*Z(C) Series MFC is selected.

The screenshot shows the 'Certification Data' tab of the AE DMFC utility tool. The window title is 'AE DMFC utility tool Rev.1.0 [HDP CVD#1 / Chamber#2 / 0 / 100sccm]'. The 'Response Check' tab is selected. The 'Flow accuracy' section contains two tables: 'Calibration Gas : N2' and 'CF Test Gas : SF6'. The 'Response' section shows settling times for 100%, 25%, and 10% setpoints. Below these are 'Zero Output' and 'Valve sheet leak' values. At the bottom are 'Write to Excel' and 'Read data' buttons. Red lines connect numbered callouts to specific elements: 1 points to the Calibration Gas table, 2 to the CF Test Gas table, 3 to the CF Error column, 4 to the Response section, 5 to the Zero Output/Valve sheet leak values, 6 to the Write to Excel button, and 8 to the Read data button.

Flow accuracy

SetPoint	Calibration Gas : N2			CF Test Gas : SF6			CF Error [%]
	Set Flow [sccm]	Measured [sccm]	Error [%RDG]	Set Flow [sccm]	Measured [sccm]	Error [%RDG]	
100%	100.00	100.21	0.212	26.50	26.59	0.342	0.13
90%	90.00	90.59	0.653				
80%	80.00	79.55	-0.564				
70%	70.00	69.84	-0.223				
60%	60.00	60.07	0.123	15.90	16.04	0.898	0.77
50%	50.00	50.00	0.004				
40%	40.00	39.97	-0.071				
30%	30.00	29.74	-0.880	7.95	7.92	-0.438	0.45
20%	20.00	19.97	-0.145				
10%	10.00	10.02	0.236				

Response

SetPoint [%]	Settling Time [sec]
100	0.823
25	0.845
10	0.867

Zero Output 0.998 [%]
Valve sheet leak 0.119 [%]

Write to Excel Read data

0%

① Flow accuracy [Calibration Gas]
② Flow accuracy [CF Test Gas]
③ Flow accuracy [CF Error]
④ Response
⑤ Zero Output Valve sheet leak
⑥ Write to Excel button
⑧ Read data button

① Flow accuracy [Calibration Gas]

This lists the record of results of calibration gas-applied calibration.

② Flow accuracy [CF Test Gas]

This lists the record of results of CF test gas-applied calibration.

③ Flow accuracy [CF Error]

This lists the CF error between calibration gas and test gas.

④ Response

This displays the ultimate time (sec) of flow with factory-default calibration gas as follows:
0→100[%]±2[%], 0→25[%]±0.5[%], and 0→10[%]±0.5[%].

⑤ Zero Output/Valve sheet leak

This displays the factory-shipped zero-point output [%] and valve leak [%].

⑥ Write to Excel button

This button is used to send calibration data, which is called up with **Read data** button, to MS Excel™.

Example: When calibration data provided on the previous page is sent to MS Excel™

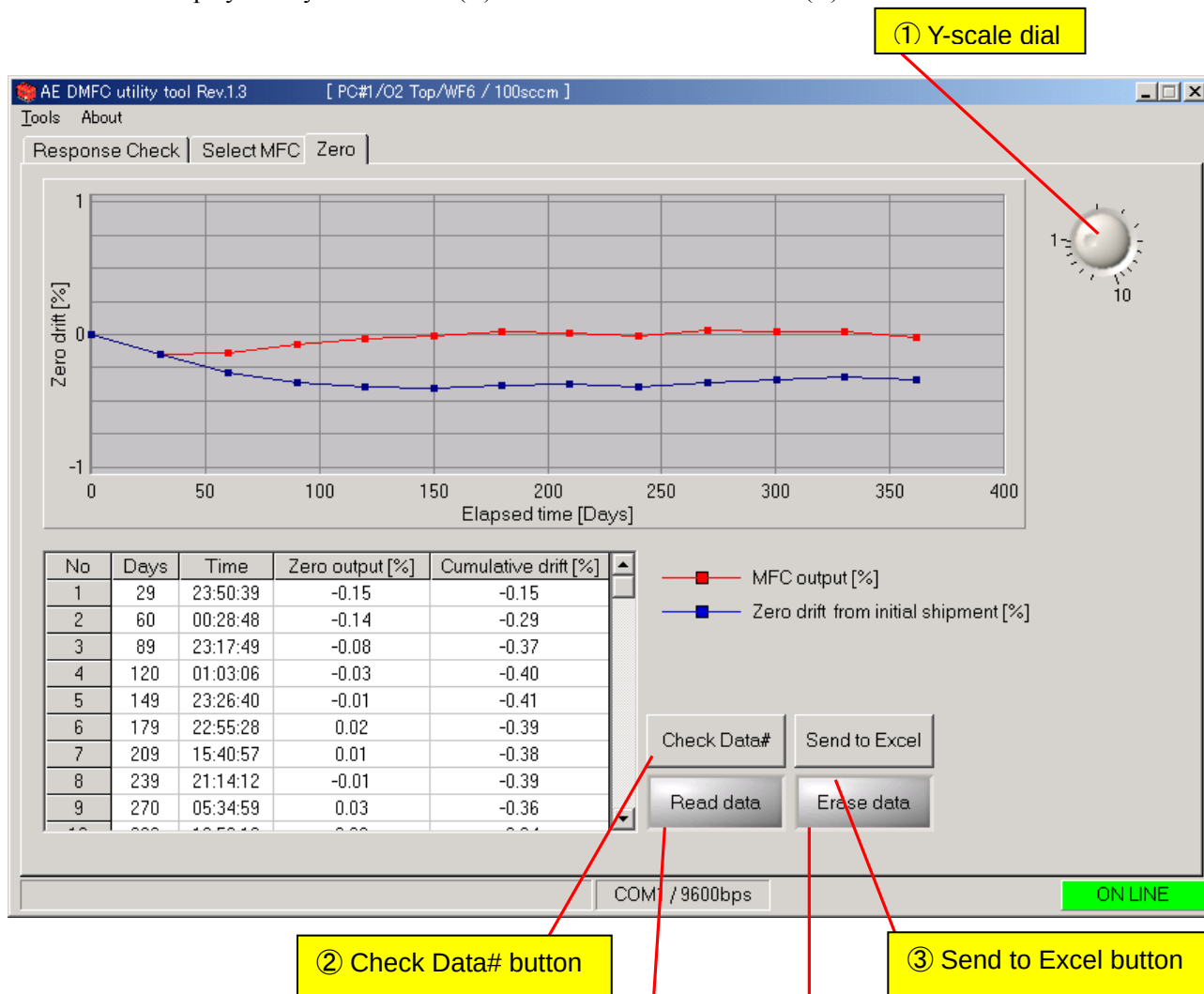
D16	fx						
A	B	C	D	E	F	G	H
	Calibration Gas : N2			CF Test Gas : SF6			
	SetFlow	Measured	Error	SetFlow	Measured	Error	CF Error
	[sccm]	[sccm]	[%RDG]	[sccm]	[sccm]	[%RDG]	[%]
100%	100	100.21	0.212	26.5	26.59	0.342	0.13
90%	90	90.59	0.653				
80%	80	79.55	-0.564				
70%	70	69.84	-0.223				
60%	60	60.07	0.123	15.9	16.04	0.898	0.77
50%	50	50	0.004				
40%	40	39.97	-0.071				
30%	30	29.74	-0.88	7.95	7.92	-0.438	0.45
20%	20	19.97	-0.145				
10%	10	10.02	0.236				
SetPoint [%]	Settling Time [sec]						
100%	0.823						
25%	0.845						
10%	0.867						
Zero output	0.998						
Valve leak	0.119						

⑦ Read data button

This button is used to call up calibration data (① to ⑤) from the active MFC.

5.5 Zero Screen

This screen is displayed only if FC-PA78*(C) Series MFC and FC-PA78*Z(C) Series MFC are selected.

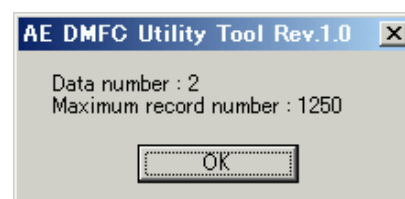


① Y-scale dial

This dial is used to change the vertical axis scale of the graph.
Scale range: ± 0.5 to ± 10 [%], expressed by 1%

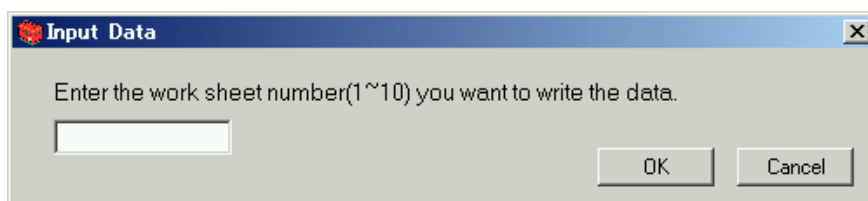
② Check Data# button

This button is used to check the number of zero-point records stored in the active MFC. Storage of up to 1250 records is permitted, and the screen shown at right provides an example of two records being stored.

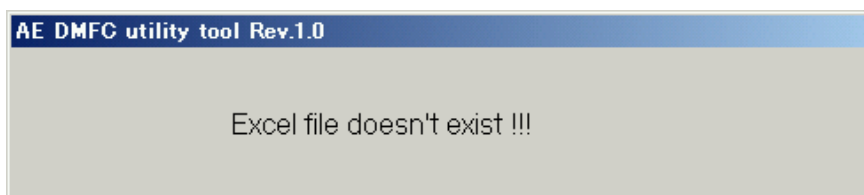


③ Send to Excel button

This button is used to send record data, which is read from the active MFC, to MS Excel™. With the click of this button, the dialog box shown below is displayed. Enter any work sheet number and click on the OK button.



The following message is displayed if MS Excel™ is not activated. In the event that the message appears, launch MS Excel™ and re-click on the Send to Excel button.

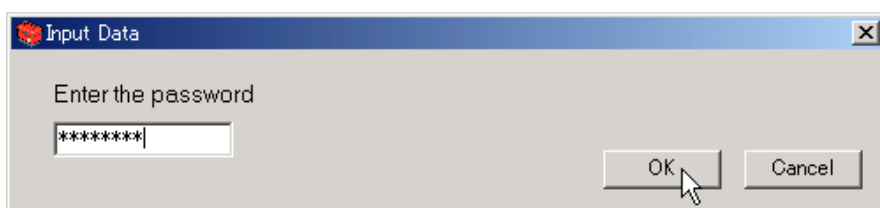


④ Read data button

This button is used to call up record data from the active MFC.

⑤ Erase data button

This button is used to delete record data from the active MFC. With the click of this button, the dialog box shown below is displayed. Enter a password.



The following dialog box is displayed if a valid password is presented to the computer.



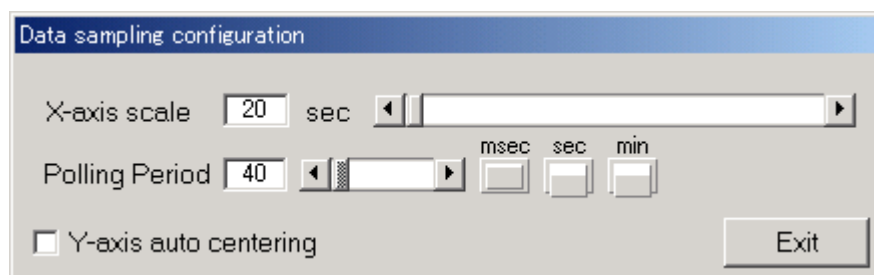
Click on **OK** button to delete data, and **Cancel** button to cancel data deletion.

6 Others

6.1 Setting of Sampling Time

This section describes the following settings and function at the time of response waveform acquisition performed from the Response Check screen: setting of sampling time, setting of time-axis scale, and Y-axis automatic centering function.

The following dialog box is displayed with the click of “Tools” in the menu bar and choosing “Plot configuration” or with the double click of any point on the graph.



X-axis scale

This allows you to change the time-axis scale via dragging the slider given at right.

Polling Period

This allows you to set the sampling time in numerical value or time.

Y-axis auto centering

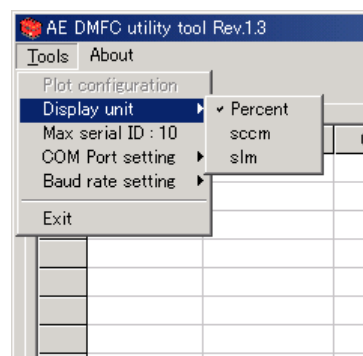
Automatic calibration of the display range is performed to keep a flow setpoint at the center of the graph on the Response screen if this checkbox is marked. The width of the analog meter is applied to that of display.

6.2 Setting of Flow Rate Display Unit

The display unit for the flow setpoint and output falls into three types: percent, sccm, and slm.

Click on “Tools” in the menu bar and choose a display unit from Display unit menu.

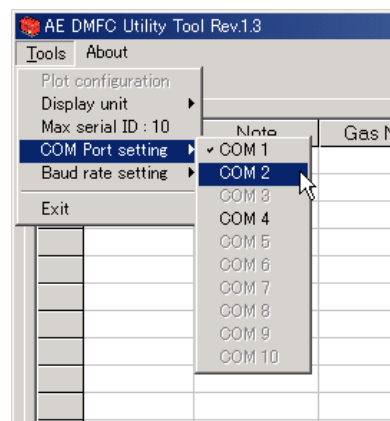
The default is “Percent” upon program start.



6.3 Setting of COM Port

COM ports with port numbers 1 to 10 assigned can be connected. This application enables available COM ports to be searched upon program start, and invalid ports are grayed out. The screen shown at right provides an example that COM1 is currently selected out of available ports, COM1, COM2 and COM4.

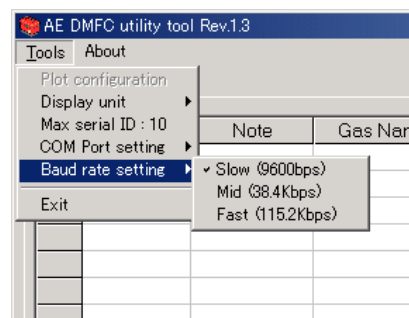
COM port setting is available only when communication with DMFC is disconnected.



6.4 Setting of Baud Rate

9600-bps communication permits sampling to be executed at speeds 80 to 100 msec, and some of FC-PA780ZC Series DMFCs are capable of supporting 115.2-Kbps communication that enables 15-msec sampling. This section describes the procedure for changing a baud rate.

- ① Click on “Tools” in the menu bar and choose “Serial port” and “BPS rate setting” to designate “Fast (115.2Kbps)” for a baud rate.
- ② This application automatically brings serial communication to tentative disconnection and allows it to be reconnected.
- ③ Make sure of no false information on MFC being provided in the list.
- ④ Click on the Serial-Connect button to disconnect serial communication.



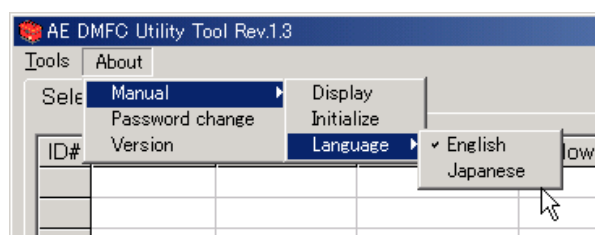
For communication at 115.2 Kbps, see “3.2 RS-232C/RS-485 Communication Converter” for converter types.

Note:

This application runs at 9600 bps of baud rate. If this application is connected to MFC with 115.2 Kbps predetermined for the baud rate, an error message is displayed that disables further connection. For sidestepping this error, click on “Tools” in the menu bar and choose “Serial port” and “BPS rate setting”. Designate “Fast (115.2Kbps)”, and then click on the Serial-Disconnect button.

6.5 Help

You can access the on-screen Instruction Manual from Display menu that is opened by clicking on “About” in the menu bar and choosing “Manual”. This function is available only on the computer with Adobe Acrobat Reader 5.0 or greater installed. The on-screen Instruction Manual is available in Japanese and English. For language selection, click on “About” in the menu bar and choose “Manual” and “Language” in sequence. (Default: English)



The Instruction Manuals are stored under the file names shown below:

Japanese version: DMFC_Utility_Tool_Manual_Japanese.pdf

English version: DMFC_Utility_Tool_Manual_English.pdf

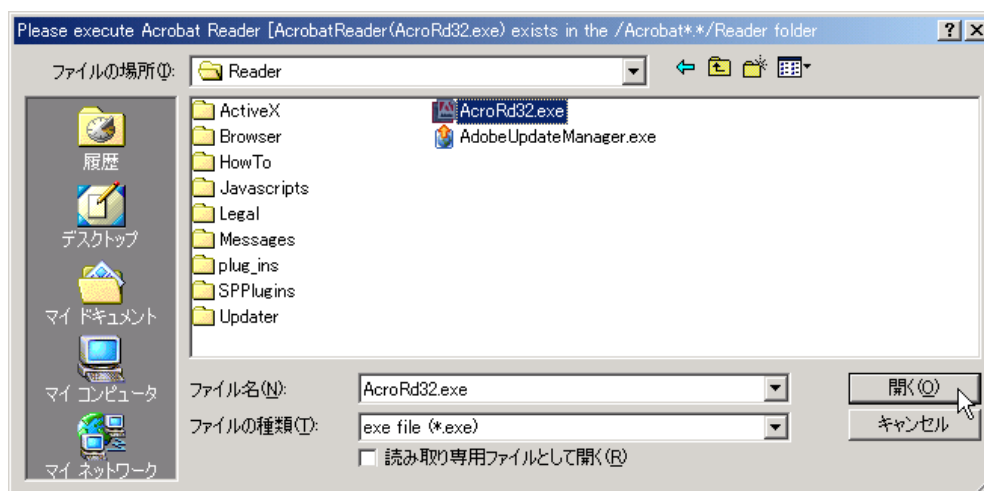
The manuals are retained in the folder that the application runs on.

The initial use of this function is to be attempted by the launch of Acrobat Reader from the dialog.

Upon installation of Acrobat Reader at the default, information is stored in an executive file (AcroRd32.exe) under the folder of C:\Program Files\Adobe\Acrobat *. *Reader.

Automatic program start is ensured in further use of this function.

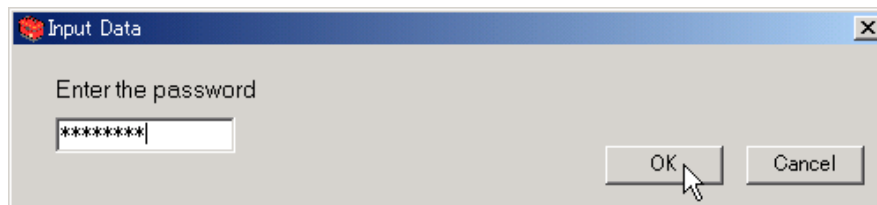
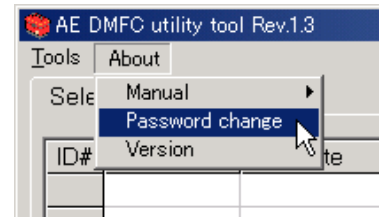
To change the folder for Acrobat Reader upgrading, go to “About” in the menu bar and choose “Manual” and “Initialize” in sequence. Initialization takes place, which urges you to restart Acrobat Reader from the dialog.



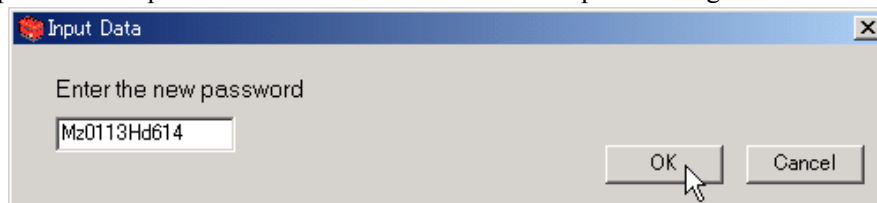
6.6 Password Change

This application is password-protected. Password change is enabled from Password change menu in “About” in the menu bar. Password entry is required for deleting zero-point data that has been stored in DMFC through the use of the zero-point record control function.

Enter the current password on the following dialog box when displayed.
Default: Password



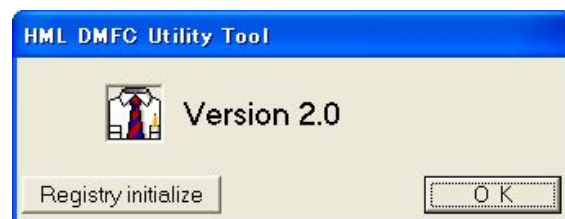
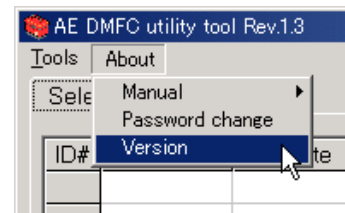
The following dialog box is displayed if agreement between input data and the current password is assured. Enter a new password upon display of the following dialog box, and re-enter the new password in response to the prompts. The new password takes effect if the re-entered password agrees.



6.7 Software Version

Software version can be checked from Version menu in “About” in the menu bar.

The Registry initialize button is used to reset software setting to its condition immediately after installation.



7 Trouble shooting

- **No recognition of DMFC at communication start**

Cause 1 : Overlap between DMFC ID numbers

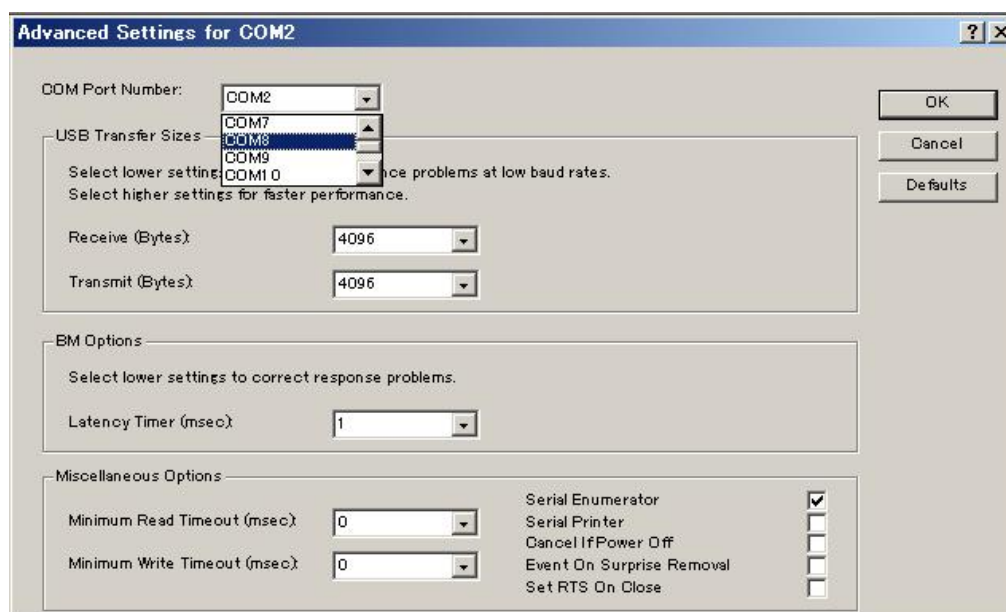
Remedy 1 : Assign a unique ID number to individual DMFC to avoid overlaps. (see section 6.2)

Cause 2 : Improper COM port number

Remedy 2 : This application is capable of detecting an available communication port number upon its launch, which enables you to select a port from “Tool” – “COM Port setting”. Not all the available ports are regarded as a RS232C communication port to capitalize on. A proper port number must be selected. If USB port connecting type USB485-AE1 is used, a COM port number is displayed in “BITS RS-485/USB Converter BSU-485/US1 (COM*)” layered under Port (COM and LPT) in Device Manager of Control Panel. In the example given below, a COM port number is set to 2.



For changing the port number, go to Property to open Port settings tab. The following window is displayed with the click of **Advanced** button, which allows changing the COM port number.



Cause 3 : Disagreement of the baud rates between MFC and software

Remedy 3 : Some of MFCs selected on the initial screen permit selection of multiple baud rates.
MFC recognition fails if the baud rate of this software disagrees with that of MFC.

Cause 4 : Inadequate Max serial ID number

Remedy 4 : DMFC ID number is to be set to 1 to 63. The maximum ID number is required to be specified in “Max serial ID”, which causes an intended MFC to fail to be detected upon communication start if the MFC ID number surpasses Max serial ID. Increment the number in Max serial ID.

Cause 5 : No communication is established if outdated version of software is used in respect of FC-D98 Series MFC.

Remedy 5 : Contact us for details.

• **Auto zero-point adjustment failed**

Cause 1 : The current zero point is out of its adjustable range.

Remedy 1 : Factory-shipped zero-point adjustment data is stored in DMFC. DMFC is configured to perform auto zero-point adjustment under the condition that comparison between the factory-shipped and the current zero-point adjustment data reveals a cumulative zero-point shift being in $\pm 10\%FS$. An error is raised if the zero-point shift is out of the adjustable range, which disables zero-point adjustment. Potential deterioration of flow accuracy in MFC may be concerned. Re-calibration of flow rate is required to recover from this error.

Cause 2 : Gas flow is observed.

Remedy 2 : Auto zero-point adjustment fails and an error is raised if gas leaks into DMFC due to incomplete shutoff of gas flow. Ensure that gas flow is stemmed completely with the shutoff valve, and retry auto zero-point adjustment.

Inquiry about Products

If you have any questions or concerns with regard to usage of our products, please contact the nearest office or our Customer Service.

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